Online Retail: Data Cleaning and Exploratory Data Analysis

AnalystLab Africa Data Analytics Internship, Week 1

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1. The Business Problem

An online retailer records every sale as a row in a transaction log, but raw logs contain missing values, duplicates, returns and pricing errors that distort revenue figures. Before the business can decide which markets, products and customers to prioritise, the data has to be cleaned and validated. This project turns 541,909 raw transaction rows into a reliable, analysis-ready dataset and uses it to answer four questions: where does revenue come from, when does it peak, which products drive demand, and how concentrated is customer spending.

2. Dataset

The Online Retail dataset holds 541,909 rows and 8 columns: InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID and Country. Transactions run from 1 December 2010 to 9 December 2011 across 38 countries.

3. Data Cleaning

Each issue was identified, quantified and resolved before analysis. The cleaning removed 17,031 rows in total (3.14%), leaving 524,878 completed sales transactions.

Issue found	Count	Action taken
Missing Description	1,454 (0.27%)	Rows removed because the product could not be identified
Missing CustomerID	135,080 (24.93%)	Retained, since they are valid sales; excluded only from customer-level analysis
Duplicate records	5,268	Removed with drop_duplicates(); re-check confirmed zero remaining
Column names	8 columns	Stripped, lowercased and underscored for consistent referencing
InvoiceDate stored as text	1 column	Converted to datetime to enable monthly trend analysis
Cancelled invoices (prefix C)	9,251	Removed because they are returns, not completed sales
Negative quantities	9,725	Removed (returns and reversals)
Zero or negative unit price	1,058	Removed (free items, adjustments or entry errors)

The cleaned file was saved as OnlineRetail_Cleaned.csv. A Revenue column (Quantity x UnitPrice) was then created to support all value-based analysis.

4. Key Insights

1. UK dependence. The United Kingdom generated about £9.00M, more than 30 times the next market (Netherlands, £285K), because the business trades overwhelmingly with domestic customers.

2. Strong seasonality. Revenue peaked at £1.50M in November 2011, nearly three times February's £523K, as holiday gifting demand built from September onwards.

3. Gifting and décor lead demand. Paper Craft, Little Birdie sold the most units (80,995) and the White Hanging Heart T-Light Holder appeared in the most orders (2,311), pointing to a gift-led product mix.

4. Concentrated customer spend. Customer 14646 spent the most (£280K) and the top 10 identified customers contributed about £1.54M, so spending is concentrated in a few high-value accounts.

5. Cleaning protected accuracy. Removing only 3.14% of rows eliminated returns, duplicates and zero-price lines that would have inflated or distorted every metric above.

5. Visual Analysis

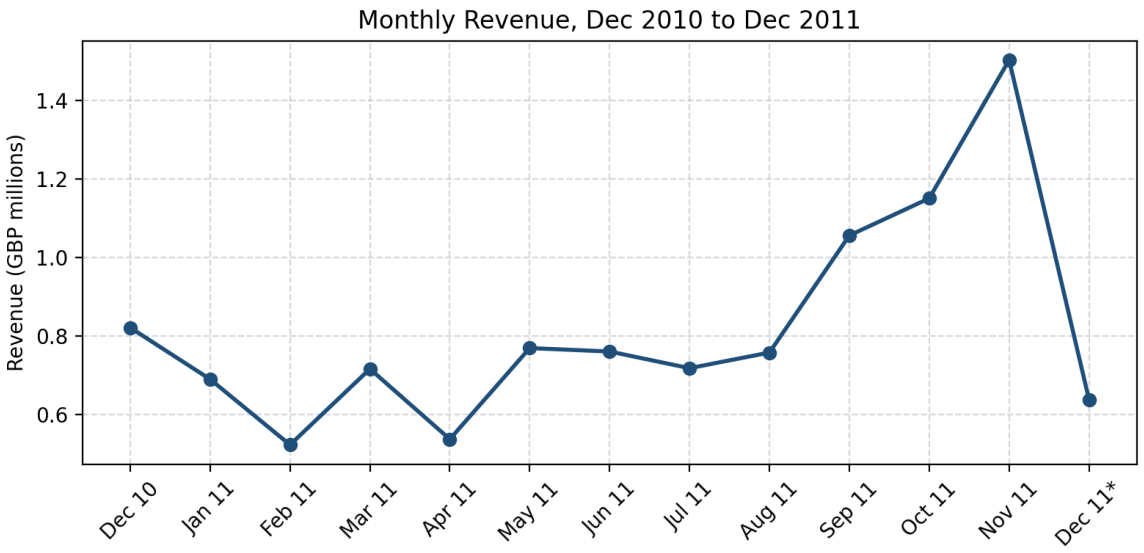


Figure 1. Monthly revenue in chronological order. *December 2011 covers only 1 to 9 December.

Insight: Revenue was uneven through mid-2011, then rose sharply from September to a November peak of about £1.50M, as holiday gifting demand built up.

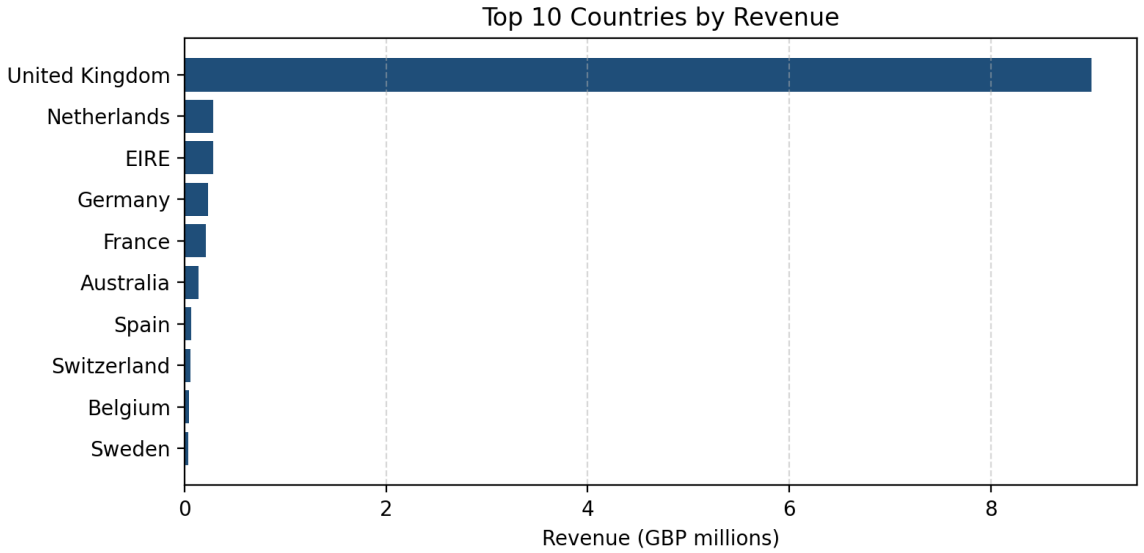


Figure 2. Top 10 countries by revenue.

Insight: The UK generated about £9.00M, over 30 times the Netherlands (£285K), because sales are overwhelmingly domestic.

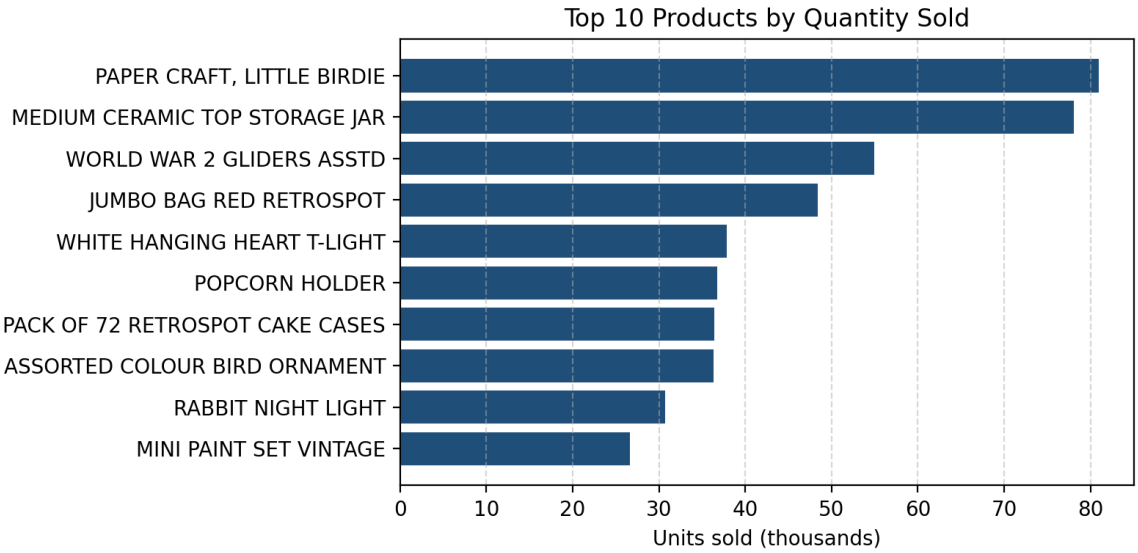


Figure 3. Best-selling products by units sold.

Insight: Paper Craft, Little Birdie led with 80,995 units, followed by Medium Ceramic Top Storage Jar (78,033), as decorative and household items dominate volume.

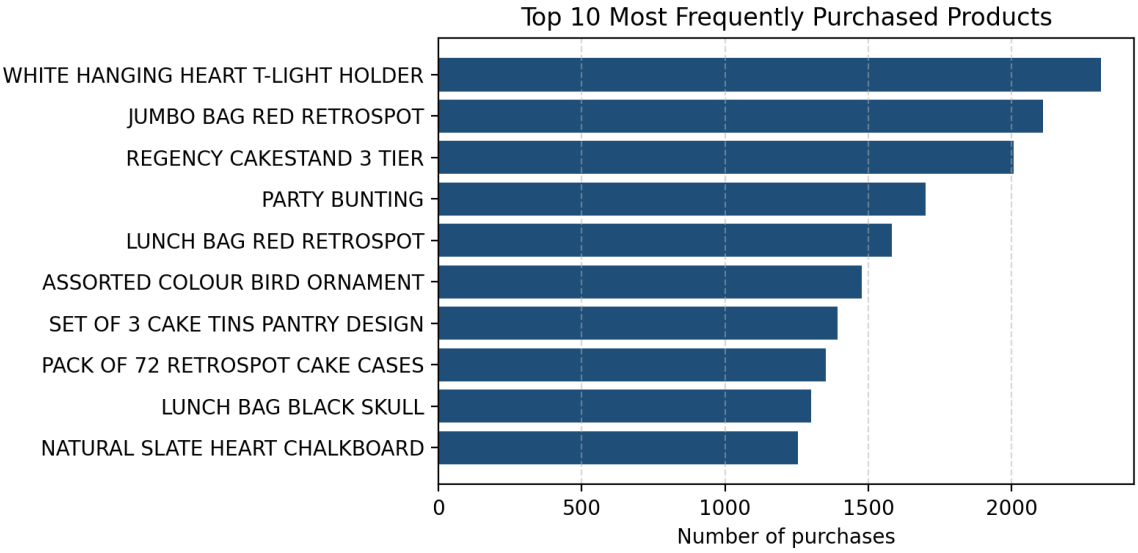


Figure 4. Products that appear in the most orders.

Insight: White Hanging Heart T-Light Holder appeared in 2,311 orders, ahead of Jumbo Bag Red Retrospot and Regency Cakestand 3 Tier, showing steady repeat demand for decorative and gift items.

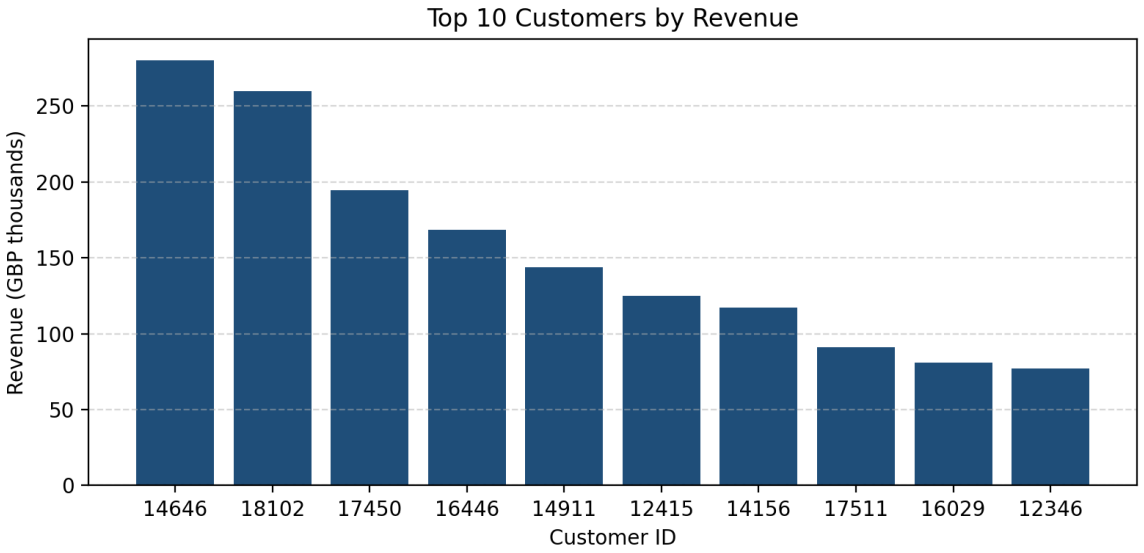


Figure 5. Top 10 customers by total revenue (customers with a recorded ID only).

Insight: Customer 14646 spent the most (about £280K) and the top 10 together contributed about £1.54M, so spending is concentrated in a few high-value accounts.

6. Recommendations

Prepare for Q4 early. Build stock and run campaigns from September, since September to November is the strongest stretch of the year.

Reduce UK reliance. Test targeted marketing in the Netherlands, Ireland, Germany and France, the strongest export markets already buying.

Protect top customers. Introduce loyalty rewards and account management for the highest-spending customers.

Prioritise stock for proven gift and décor lines. Keep best-sellers and high-frequency items well stocked and bundle them for gifting.

Capture customer IDs. About 25% of transactions have no CustomerID, which limits customer analysis. Make ID capture mandatory at checkout.

7. Limitations and Next Steps

Cancelled invoices were removed without being matched to their original orders, so a large order that was later reversed may still be counted as a sale. Matching cancellations to originals would give a stricter net-revenue view and should be checked for the very largest orders. December 2011 is a partial month. Next steps include customer segmentation (RFM), a product-level revenue analysis and an interactive dashboard.